

Software Engineer

Experienced Software Engineer who enjoys hands-on technical roles.

Technical Experience

Technology

- Languages: C, C++, Python, SQL
- Tools: MySQL, Ansible, uWSGI, Prometheus, Grafana, HCVault

Select achievements

- Pub/Sub Streaming websocket server that handles 4K events per cpu.
- Automated validation of all API input and output via openAPI specifications.
- Zero-downtime scale up of our single database/single table to partitioned databases per customer, 250 tables, 1TB.

INRIA, 2025 – Present, Research Engineer

How easy and cheap is it to build domain-specific SLMs (Small Language Models):

- Built training and logging framework for SLMs
- Trained end-to-end SLMs up to 3B parameter size
- Studied the reasons for failed training convergence

Hive, 2023 – 2025, Architect

Scaled (100x) and improved the reliability (2x) of a go libp2p-based application.

- Metric collection, analysis, monitoring, and alerting spanning centralized infrastructure and a p2p network
- Debugged and fixed network and IO problems to scale the Hive p2p network
- Built python-based development, testing, and debugging tools
- Planned and managed component-level testing strategies
- Designed forward error correction policies around dynamic decisions to minimize both data loss and storage use

Alcmeon, 2011 – 2023, CTO, co-founder

As CTO, I

- Made sure our customer's data stayed safe via at-rest encryption, and, centralized access control
- Scaled our code and infrastructure to sustain 50% YoY volume growth with 1000% YoY peak volume growth.
- Recruited and led the technical team (8 years)
- Grew and directly managed the engineering manager and product director.

As co-founder, I focused on

- Delivering high quality customer support
- Product design to ensure we build features that grow our business
- Presales to help close deals with prospects

Select projects

- Pub/Sub Streaming websocket server that scales to 10K concurrent connections per cpu
- Zero-downtime scale up of a single database with a single table to partitioned databases per customer, 250 tables, 1TB
- SSO integration in SSH
- Designed and implemented a security policy spanning infrastructure and application-level code

INRIA, 2005 – 2011, Software Lead ns-3

Built ns-3, an open-source network simulator now used in hundreds of research publications every year.

- Designed and implemented core APIs: object model, network packets, event scheduler
- Implemented models for UDP/IP/ICMP, MAC/PHY Wi-Fi network protocols
- Integrated unmodified real-world network protocol implementations

But also:

- Recruited, and managed local development team (5).
- Relocated to University of Washington for 10 months to initiate collaboration with US team.
- Evangelized use of ns-3 within other research institutes via presentations and seminars.
- Published as main author (4) and co-author (3) papers on ns-3.

INRIA, 2003 – 2005, Software Engineer

Designed and built software for network research teams:

- Yans, a C++ event-driven simulator,
- NEPI, a python tool used to describe, deploy, and control networking experiments on hundreds of hosts distributed all over the planet.

Provided software mentoring to research projects involved in bio-reactor chemical reaction control and medical image analysis.

Sigma-Designs, 2001 – 2003, Software Engineer

Implemented DVD navigation control software for the video decompression chips that were developed by Sigma-Designs and sold to OEMs to build consumer DVD players.

Education

2006 – 2010	Ph.D. at University of Nice, <i>Experimentation Tools for Networking Research</i> , under supervision from Walid Dabbous
1998 – 2001	Engineer at Telecom ParisTech (ENST), Software Engineering, Networking, Micro-Electronics